

software development processes, enabling developers to identify defects in the code very early in the development process. By ensuring compliance to a coding standard, static analysis not only identifies defects that can cause program crashes but also provides best practice, enabling the developer to reach higher level of standards, and thus prevents introduction of defects.

Static analysis automates code reviews, removing the chances of human error in the process whilst ensuring a deterministic result that removes the emotion often associated with individuals reviewing another developer's code.

There are also some static analysis tools that offer developers immediate feedback on the code they have written, allowing them to address issues before committing changes to the organisation's repository of code. These tools can also be used to provide a measure of overall project quality when used in conjunction with a software quality management system.

How fast can these static tools detect errors?

There was an interesting case study of one of our prospects working in industrial automation space and developing control instrumentation products. They were not using any static analysis tool and were mainly relying on manual code inspection. They gave a beta release of their product to one of their existing customers for testing. There was a serious bug in the code, which they were unable to find for

almost six months. We were called on-site, and within 30 minutes we were able to set up and configure the tool for their environment and identify the defect.

What are the guidelines set by IEC61508 standard?

The IEC 61508 standard includes guidance that helps developers mitigate risks or systematic failures and random hardware failures through provision of appropriate requirements and processes. With an extensive hazard analysis of not only the resulting software but also the processes used during the development cycle, these risks should be addressed.

Additional challenges include peer reviews at each decision phase and documenting bidirectional traceability of requirements through design and testing to release. On the software part, the international standard IEC 61508 mandates use of better development processes, including use of coding standards to encourage further improvements in software quality.

What languages other than 'C' and 'C++' are emerging for code development?

'C' is the most widely used language, followed by 'C++'. 'C++' is actually growing in the embedded market and also in the aviation industry. We are also looking at 'C#' as some of our clients are using it. ●

(The complete interview can be viewed at electronicsforu.com)

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